

①

Double Integrals

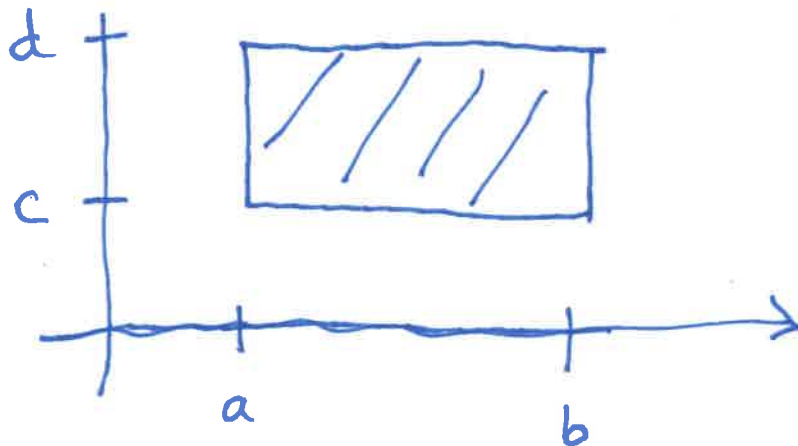
$$a < b, \quad c < d$$

$$R = [a, b] \times [c, d]$$

$$f: R \rightarrow \mathbb{R} \quad f \text{ continuous}$$

$$\iint_R f(x, y) dA$$

Subdivide into
Small rectangles



$$\Delta x = \frac{b-a}{m}$$

$$\Delta y = \frac{d-c}{n}$$

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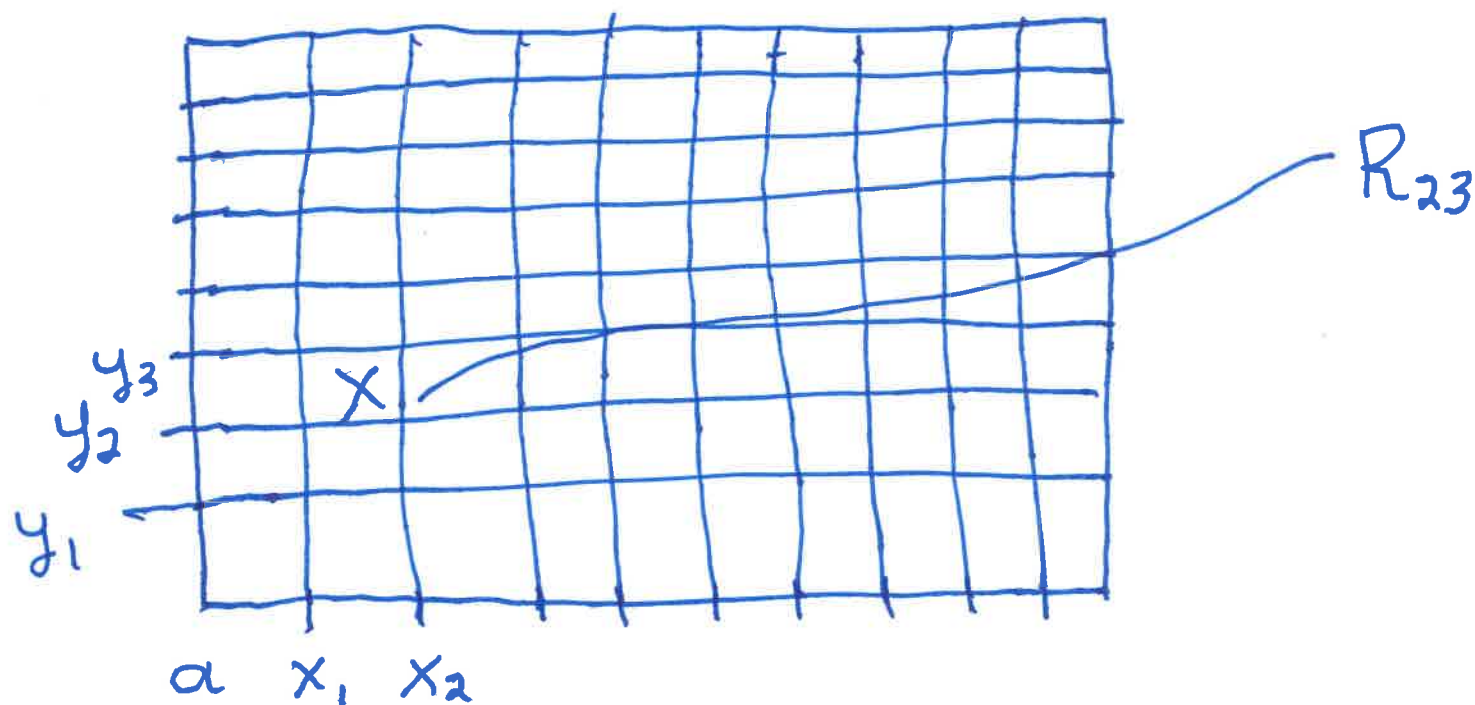
$$x_i = a + i\Delta x$$

$$i = 0, 1, 2, \dots, m$$

$$y_j = c + j\Delta y$$

$$j = 0, 1, 2, \dots, n$$

$$R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j]$$



$$I = \{ (i, j) : i = 1, 2, \dots, m, j = 1, 2, \dots, n \} \quad (3)$$

$$(x_{ij}^*, y_{ij}^*) \in R_{ij}$$

$$\iint_A f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{(i, j) \in I} f(x_{ij}^*, y_{ij}^*) \Delta x \Delta y$$

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Fubini's Theorem: If f is cont.
then

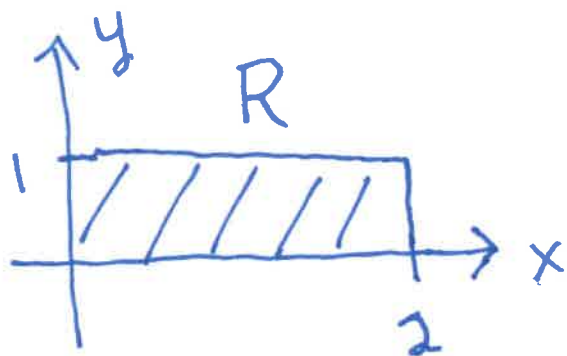
$$\int\int_R f(x,y) dA = \int_c^d \left(\int_a^b f(x,y) dx \right) dy$$

Double
Integral

$$= \int_a^b \left(\int_c^d f(x,y) dy \right) dx$$

Iterated
Integrals

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Ex:

$$f(x, y) = x^2 e^{3y}$$

$$\iint_R f(x, y) dA = \int_0^1 \left(\int_0^2 f(x, y) dx \right) dy$$

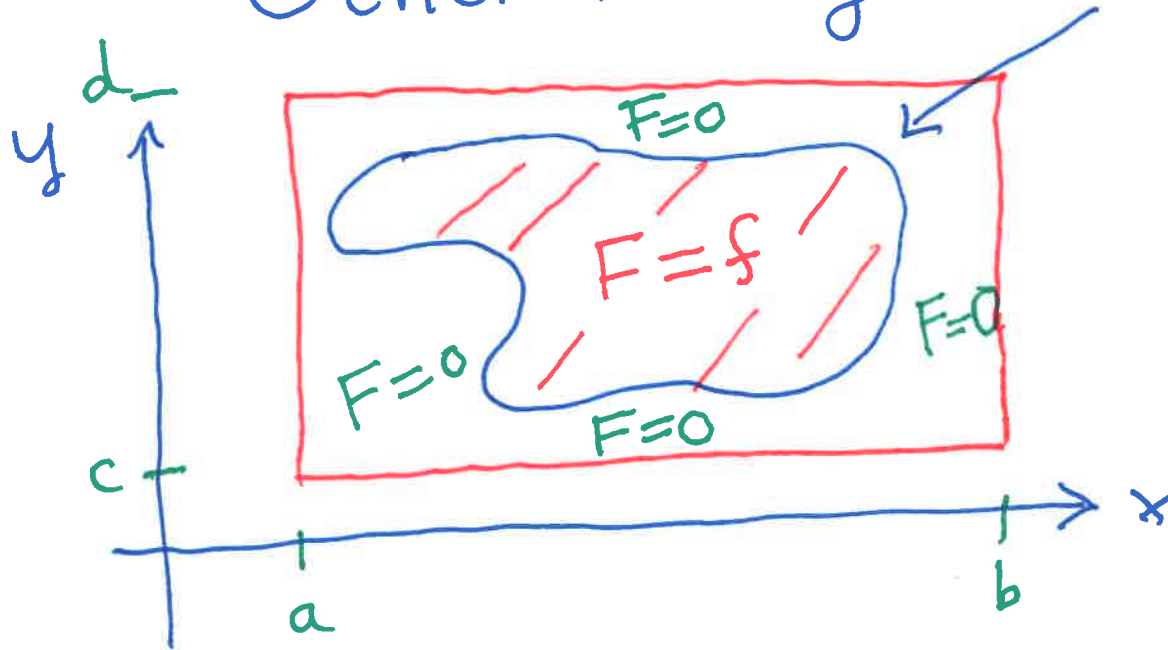
$$= \int_0^1 \int_0^2 x^2 e^{3y} dx dy$$

$$= \int_0^1 \left. \frac{1}{3} x^3 e^{3y} \right|_{x=0}^2 dy = \int_0^1 \left(\frac{8}{3} e^{3y} - 0 \right) dy$$

$$= \frac{8}{9} e^{3y} \Big|_0^1 = \frac{8}{9} (e^3 - 1)$$

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General Regions



Even with really small rectangles
won't get an exact fit

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Choose $[a,b] \times [c,d]$ that contains R

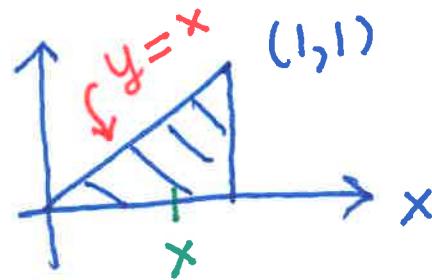
$$F(x,y) = \begin{cases} f(x,y), & (x,y) \in R \\ 0 & \text{otherwise} \end{cases}$$

Define $\iint_R f(x,y) dA = \iint_{[a,b] \times [c,d]} F(x,y) dA$

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Miracle: This works!

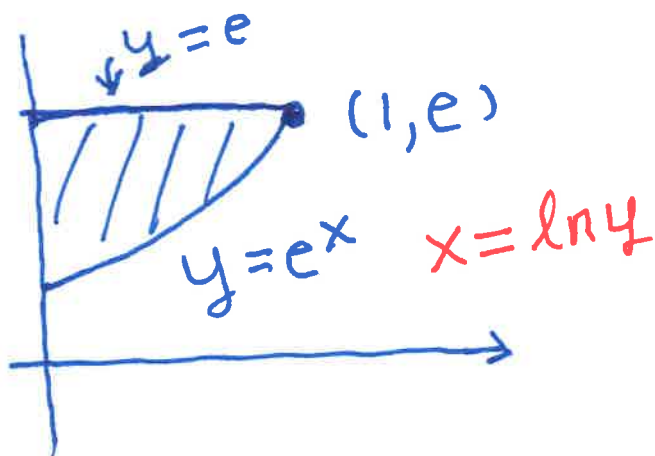
Ex: a)



$$\iint_R f(x,y) dA = \int_0^1 \int_0^x f(x,y) dy dx$$

$$= \int_0^1 \int_y^1 f(x,y) dx dy$$

b)



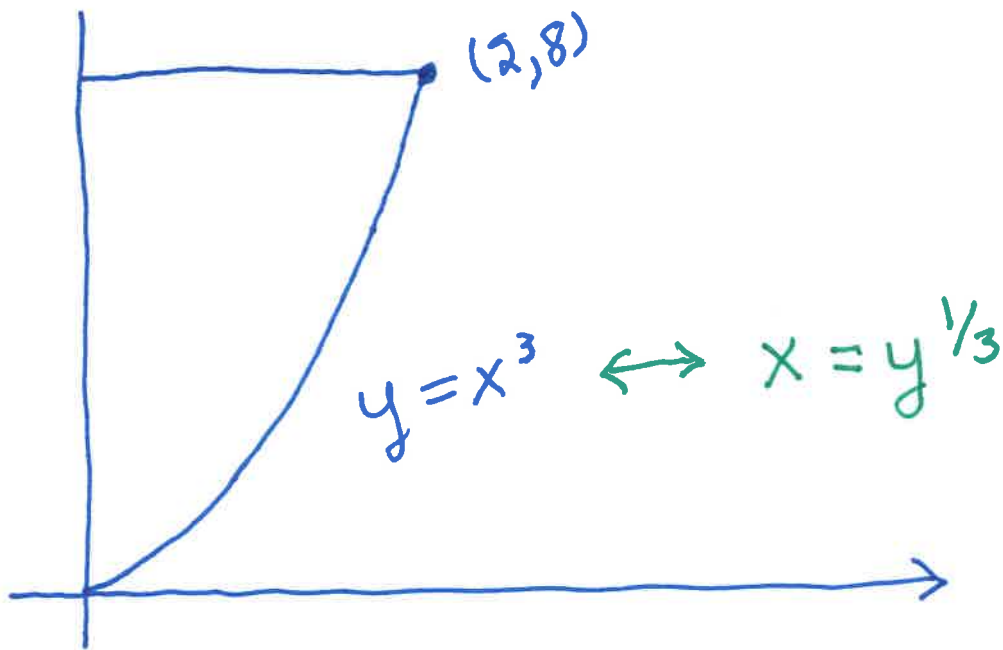
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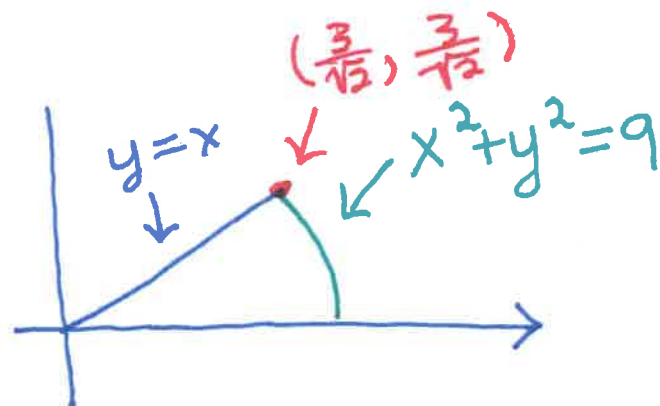
$$\begin{aligned}\iint_R f(x, y) dA &= \int_0^1 \int_{e^x}^e f(x, y) dy dx \\ &= \int_1^e \int_0^{\ln y} f(x, y) dx dy\end{aligned}$$

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Ex: Reverse order of Integration

$$\int_0^2 \int_{x^3}^8 f(x,y) dy dx = \int_0^8 \int_0^{y^{1/3}} f(x,y) dx dy$$



Ex:

$$x = \sqrt{9-y^2}$$

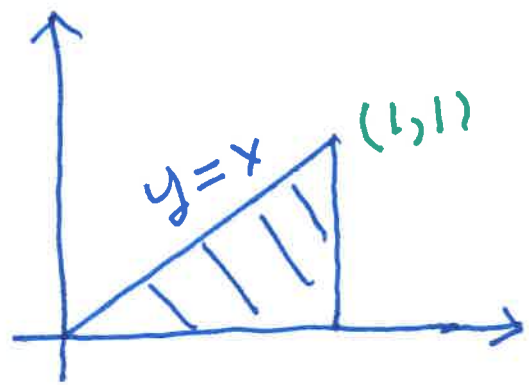
$$y = \sqrt{9-x^2}$$

$$\iint_R f(x,y) dA = \int_0^{\frac{3}{\sqrt{2}}} \int_y^{\sqrt{9-y^2}} f(x,y) dx dy$$

$$= \int_0^{\frac{3}{\sqrt{2}}} \int_0^x f(x,y) dy dx + \int_{\frac{3}{\sqrt{2}}}^3 \int_0^{\sqrt{9-x^2}} f(x,y) dy dx$$

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Ex: $\int_0^1 \int_y^1 e^{x^2} dx dy$

Reverse
Order

$$= \int_0^1 \int_0^x e^{x^2} dy dx$$

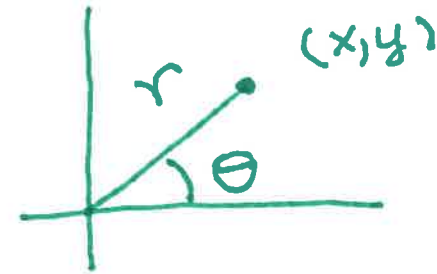
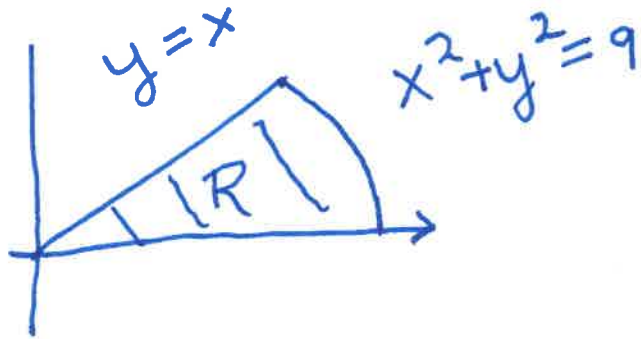
$$= \int_0^1 e^{x^2} y \Big|_{y=0}^x dx$$

$$= \int_0^1 x e^{x^2} dx \quad (\text{2nd Miracle})$$

$$= \frac{1}{2} e^{x^2} \Big|_0^1 = \frac{1}{2} (e - 1)$$

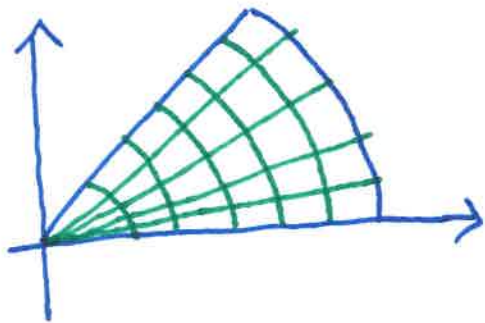
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Polar Coords



$$x = r \cos \theta$$
$$y = r \sin \theta$$

In Polar : $0 \leq r \leq 3$
 $0 \leq \theta \leq \frac{\pi}{4}$



$$r \Delta \theta \begin{matrix} \Delta r \\ \Delta r \end{matrix} \approx r \Delta \theta$$
$$[(r + \Delta r) \Delta \theta]$$

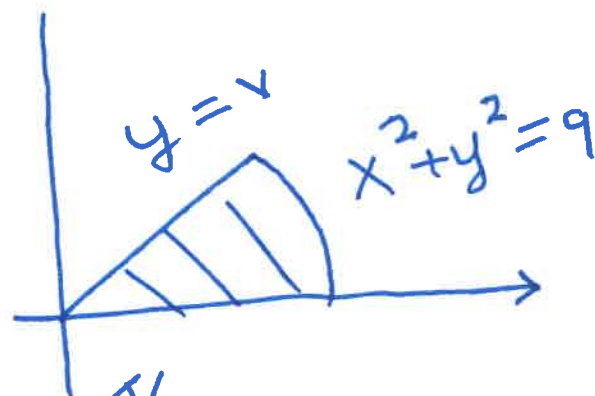
$$\Delta A \approx r \Delta r \Delta \theta$$

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$$dA = r dr d\theta$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$



$$\int_R \underbrace{\sqrt{x^2 + y^2}}_{=r} dA = \int_0^{\pi/4} \int_0^3 r (r dr d\theta) \quad \begin{matrix} \nearrow \\ dA \end{matrix}$$

$$= \int_0^{\pi/4} \int_0^3 r^2 dr d\theta = \int_0^{\pi/4} \left. \frac{1}{3} r^3 \right|_0^3 d\theta$$

$$= \int_0^{\pi/4} 9 d\theta = 9\theta \Big|_0^{\pi/4} = \frac{9\pi}{4}$$

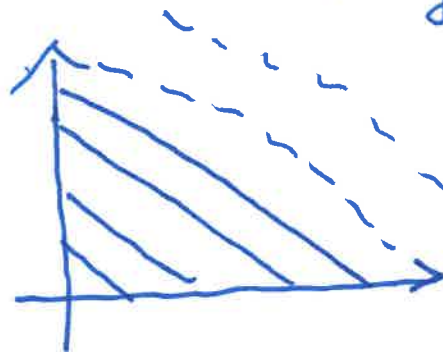
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Calculate $I = \int_0^{\infty} e^{-x^2} dx$

$$I = \int_0^{\infty} e^{-y^2} dy$$

$$I^2 = \left(\int_0^{\infty} e^{-x^2} dx \right) \left(\int_0^{\infty} e^{-y^2} dy \right)$$

$$= \int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} dy dx$$



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Switch To Polar

$$I^2 = \int_0^{\pi/2} \int_0^{\infty} e^{-r^2} r dr d\theta \quad (\text{Miracle})$$

$$= \int_0^{\pi/2} -\frac{1}{2} e^{-r^2} \Big|_0^{\infty} d\theta = \int_0^{\pi/2} \frac{1}{2} d\theta$$

$$= \frac{\pi}{4} \quad I = \frac{\sqrt{\pi}}{2}$$

Note $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$

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Black-Scholes Calculations

Single Stock , No Dividends

Bank Acct r - constant

Continuous Compounding (Risk-Free)

σ - Volatility (annualized)

S_t - Stock Price at t

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European Style Deriv. Security
Single Payment of $g(S_T)$
at time T .

Distribution of $\ln(S_T)$ under
risk-neutral measure (pricing
measure) is Normal with

Mean $\ln(S_0) + (r - \frac{1}{2}\sigma^2)T$

Variance v^2 , $v = \sigma\sqrt{T}$

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Option V

$$V_T = g(S_T)$$

Expectation
under risk-
neutral meas.

Time-0 Price

$$V_0 = e^{-rT} \tilde{E}[g(S_T)]$$

$$S_T = e^{\underline{X}}$$

X normal

Mean $\mu = \ln S_0 + (r - \frac{1}{2}\sigma^2)T$

Sd. Dev ν , $\nu^2 = \sigma^2 T$

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$$V_0 = e^{-rT} \int_{-\infty}^{\infty} g(e^x) \psi(x) dx$$

ψ - Density Normal Mean μ
Var σ^2

ϕ - Std Normal Density

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

$$\Phi(x) = \int_{-\infty}^x \phi(t) dt$$

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$$\begin{aligned}\psi(x) &= \frac{1}{\sigma} g\left(\frac{x-\mu}{\sigma}\right) \\ &= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}\end{aligned}$$

European Put

$$g(x) = \begin{cases} K-x & x \leq K \\ 0 & x > K \end{cases}$$

(Sell share for K at T if
you want)

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$$P_T = \begin{cases} K - S_T, & S_T \leq K \\ 0, & S_T > K \end{cases}$$

Pos. Part $x^+ = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases}$

$$P_T = (K - S_T)^+$$

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$$P_0 = e^{-rT} \int_{-\infty}^{\infty} (k - e^x)^+ \frac{1}{\sigma} \varphi\left(\frac{x-\mu}{\sigma}\right) dx$$

∴ Get Rid of + in $(k - e^x)^+$

$$= e^{-rT} \int_{-\infty}^{\ln k} (k - e^x) \frac{1}{\sigma} \varphi\left(\frac{x-\mu}{\sigma}\right) dx$$

$$(k - e^x)^+ = 0, \quad x \geq \ln k$$

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Change Variables

$$\text{Let } y = \frac{x - \mu}{\sigma}$$

$$x = \mu + \sigma y$$

$$dx = \sigma dy$$

$$P_0 = e^{-rT} \int_{-\infty}^c (\kappa - e^{\mu + \sigma y}) g(y) dy$$

$$c = \frac{\ln \kappa - \mu}{\sigma}$$

$$X \rightarrow -\infty$$

$$X = \ln \kappa$$

$$y \rightarrow -\infty$$

$$y = \frac{\ln \kappa - \mu}{\sigma}$$

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$$P_0 = e^{-rT} K \int_{-\infty}^c g(y) dy$$

$$- e^{-rT} \int_{-\infty}^c e^{\mu + \nu y} g(y) dy$$

$$= e^{-rT} K \Phi(c) - e^{-rT} I$$

$$I = \int_{-\infty}^c e^{\mu + \nu y} g(y) dy$$

Calculate I

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$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^c e^{\mu + \nu y} e^{-\frac{1}{2}y^2} dy$$

$$= \frac{e^{\mu}}{\sqrt{2\pi}} \int_{-\infty}^c e^{\nu y - \frac{1}{2}y^2} dy$$

Complete the Square

$$\nu y - \frac{1}{2}y^2 = -\frac{1}{2}(y - \nu)^2 + \frac{1}{2}\nu^2$$

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$$I = \frac{e^{\mu}}{\sqrt{2\pi}} \int_{-\infty}^c e^{\frac{1}{2}v^2} e^{-\frac{1}{2}(y-v)^2} dy$$

$$= e^{\mu + \frac{1}{2}v^2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^c e^{-\frac{1}{2}(y-v)^2} dy$$


Let $z = y - v$, $dz = dy$

$$= e^{\mu + \frac{1}{2}v^2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{c-v} e^{-\frac{1}{2}z^2} dz$$

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$$I = e^{\mu + \frac{1}{2}v^2} \Phi(c-v)$$

$$P_0 = e^{-rT} K \Phi(c) - e^{-rT} e^{\mu + \frac{1}{2}v^2} \Phi(c-v)$$

$$\mu + \frac{1}{2}v^2 = \ln S_0 + rT$$

$$P_0 = e^{-rT} K \Phi(c) - S_0 \Phi(c-v)$$

$$v = \sigma \sqrt{T}$$

$$c = \frac{\ln K - \mu}{v}$$